Engineering Project
Final Deliverable

ENGR 110

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Problem Definition

In this project, we will be focusing on medicine mismanagement problem which most older people are dealing with.

Need

The reason we focus on the Medicine Mismanagement as a key issue is that many older adults struggle to remember when to take their medications, often leading to missed doses or accidental overdosing. Medicine mismanagement can have major consequences, affecting both health and the ability of older adults to live independently. This need was selected because a solution that helps older adults managing medications safely would reduce health risks and improve quality of life. The proposed solution is a system that reminds older adults to take the correct dose at the right time, promoting independence and offering peace of mind for caregivers.

Possible Clients

- **BC Seniors Advocate**
    - Focuses on improving the quality of life and safety of seniors.
    - Likely interested in solutions that promote medication adherence and independence.

- **Senior Health Clinics**
    - Provide direct care to older adults and aim to reduce health risks.
    - Can implement the system as part of their patient care services.

- **Healthcare Providers Serving Older Adults**
    - Invested in enhancing patient safety and reducing medication-related complications.
    - May use the system to support caregivers and streamline patient management.

The Goal

The goal is to create a system to address medication mismanagement in the elderly, focusing on organizing both the volume and timing of medication doses. The solution should consider the varying doses and schedules for different medications. A key aspect is how seniors will set up the initial plan and stock their medication.Due to difficulties seniors often

face with complex software interfaces, the system should be simple, or ideally, avoid computer-based solutions. Additionally, it would be helpful for the system to be set up by pharmacists or doctors to ensure proper scheduling and avoid errors in medication management.

- o **Organize medication** doses with respect to both volume and timing.
- o **Account for varying doses and schedules** for multiple medications.
- o **Simple setup** process for seniors, avoiding complex software interfaces.
- o **Professional setup** by pharmacists or doctors to ensure accurate scheduling.

Research

There are both positives and drawbacks. Smart pill dispensers and reminder apps offer organization and reminders but can be costly or hard for seniors to use. Blister packs from pharmacies make dosing easy but lack reminders.

In order to understand the problem better, we think the best questions to as are:

- o What are the main reasons older people forget their medications (memory issues or complex schedules?
- o What level of technology are the seniors in BC comfortable with to help us decide on an app or device?
- o Should the system notify caregivers if a dose is missed?

Causes of medication mismanagement and known solutions

- **Cause**
  - o Patients with dementia often misuse their medicine due to their impaired decision-making capacity, poorer adherence, and communication challenges.[1]
  - o Medicine Prescriptions are too complex and hard to understand but easy to forget. [1]
  - o Too much unnecessary medicines makes the patient confused.[1]
  - o Miscommunication between healthcare institutes during transition in care.[1]

- **Solution**
  - o Make the prescriptions much easier for the patients with dementia to

understand. [1]

o Use technology to remind the patients about their prescriptions and when they should take their medicine.[1]

Positive and negative aspects of existing designs

In order to properly determine good design attributes, we identified the positive and negative aspects of existing solutions found in our research.

- **Positives**

  o Upfront cost instead of monthly payments to focus on long term affordability for seniors [2], which is important considering the financial difficulties seniors face. [possible reference for financial difficulty amongst the elderly?]

  o Up to date user interface to ensure smooth experience and reduce confusion [2].

  o Display for ease of use [2]

  o Mobile app integration allows for remote monitoring and easy set up [2].

  o Ability to contact help when detecting an over or underdose [3].

  o Tracks hand movement to ensure proper use of dispenser [3].

  o Technology and online tools provide flexible ways to handle problems [4].

  o The method of making the medical prescription less complicated is simple and affordable [1].

  o Using technology to involve in medicine management doesn't rely much on the patient's cognitive ability [1].

- **Negatives**

  o Has multiple limitations that affect the effectiveness of the method of making the medical prescription less complicated [1].

  o Limited number of dosage(s) per day.

  o Limited range of dementia severity.

  o The method of applying technology into medicine management is costly [1].

  o Some technological solutions are too complex for older adults to handle [4].

  o Older people may become overly dependent on technology. If it fails to function, they could miss doses and experience health risk.[1]

  o Some design may need regular software update which could be confusing for the elder people.[2]

  o The price for technologies can be high, which could be difficult for some older adults to afford [4].

  o The method of making the prescription less complicated requires more

involvement from the caregivers [1].

o Reliance on significant background software and hardware may cause problems with resilience and repair.[2][3]

Design Objectives and constraints

- **Objectives**

  o The device must be easily restocked.

  o The device must be easy to use.

  o The device must be compact.

  o The device must be affordable.

  o The device must remind the patient to take their medications.

  o The device must be able to store a certain number of medications.

  o The device must properly dose the medications.

  o The device must be easy to clean.

- **Constraints**

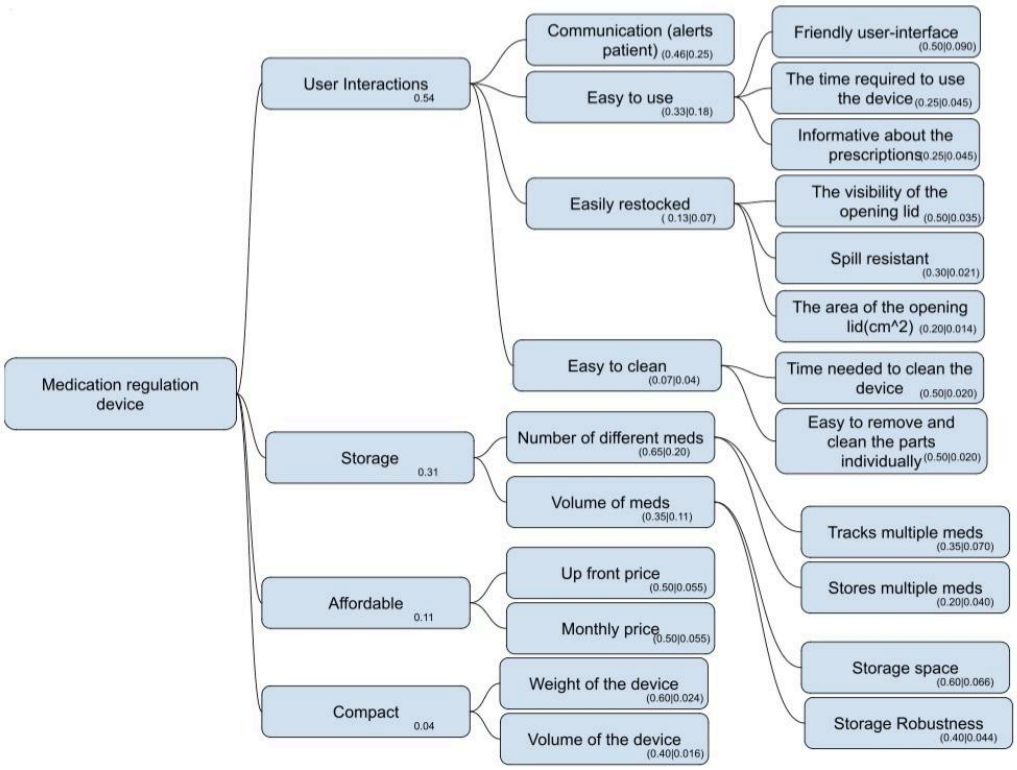
  o The device must be child-proof, should not be able to be unlocked by children or should only be able to be unlocked by permitted people.

  o The device must be food safe, the materials used in construction must be sanitary and approved for food use.

  o The device must preserve the medications, watertight to keep out humidity and enclosed to keep out sunlight or any other external impacts on meds.

  o The device needs to be safe; any motors or other components should not be able to harm the user (eg. trapped fingers, electrical accidents)

Weighted Criteria Tree



Problem Definition Statement

Many elderly individuals encounter significant challenges in managing their medications due to issues such as memory loss, complex medication schedules and physical limitations. These challenges cause missing doses, incorrect usage, and misuse of prescription, which decrease the effectiveness of the medications they are taking and pose serious health risks to many older adults who live alone independently. This difficulty in managing medication routines not only reduces the effectiveness of treatments but also increases health risks, especially for those who live alone and have no daily support. Without an accessible and reliable system to assist with medication management, elderly individuals may struggle to maintain their health and independence, ultimately impacting their quality of life. Additionally, poor medication adherence can lead to higher healthcare costs and increased strain on healthcare resources. Addressing this gap could bring significant value by supporting the well-being and autonomy of older adults while contributing positively to society by reducing healthcare burdens. The problem is that older adults, particularly those with memory or physical challenges, lack a reliable, accessible way to manage their medications effectively, which compromises their health, independence, and quality of life, and places additional strain on healthcare systems.

Conceptualization

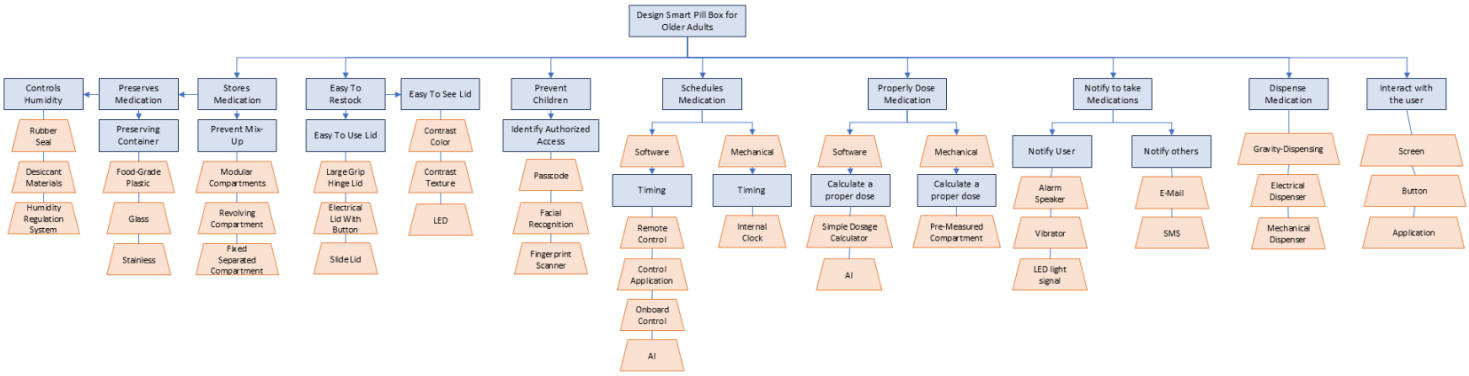
In this part, we will cover the topic of functions, conceptual designs (Morphological Charts), and performance specifications.

Functions

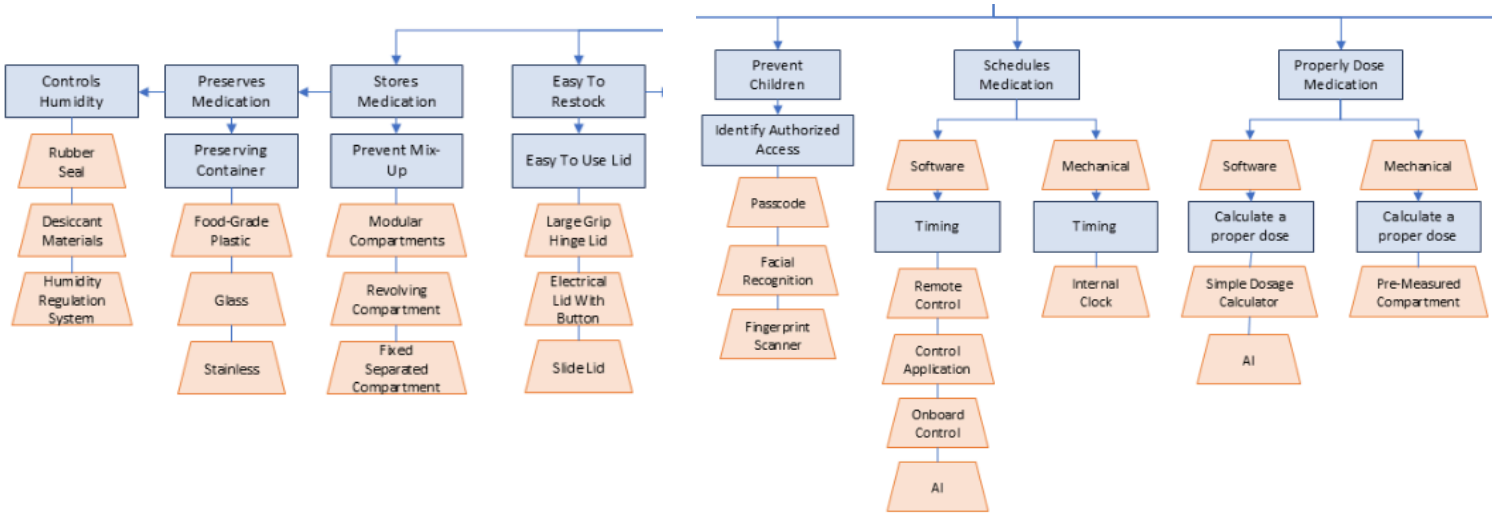
- **Stores Medication**
  The device must be able to store medication and preserve the medications.
  - **Control Humidity**
    The device must be able to regulate humidity to preserve the medications for longer use.
  - **Preserving Container**
    The device must be made from a material that promotes medication preservation for safety in consumption.
  - **Prevent Mix-Up**
    The device must be able to store different types of medications and prevent Mix-Up within its storage to ensure safety in consumption.
- **Easy to use**
  The device must be designed to be easy to use by the senior patient.
  - **Easy to Use Lid**
    The device's lid must be easy to use by the senior patients to ensure that they can use the device independently.
  - **Easy to See Lid**
    The device's lid must be visible for the senior patients to ensure the ease of use for seniors with conditions like impaired vision.
- **Prevent Children**
  The device must be able to prevent access from children to prevent accidental consumption by childrens.
  - **Prevent Unauthorized Access**
    The device must be able to prevent unauthorized access from the unauthorized individuals to ensure security and prevent possible risks from unauthorized access poisoning, intended medication mix-up, etc.
- **Schedules Medication**
  The device must be able to schedule medication accurately to assist the senior patients.
- **Properly Doses Medication**
  The device must be able to dose medication to the senior patient correctly by calculating the amount of medications needed per consumption.
- **Notify to Take Medications**
  The device must be able to notify the senior patient when to take their medication to provide maximum effectiveness of medication treatments.
- **Dispense Medications**
  The device must be able to dispense medication to the senior patient
- **Interact with the user**
  The device must have easy-to-use user interface for seniors.

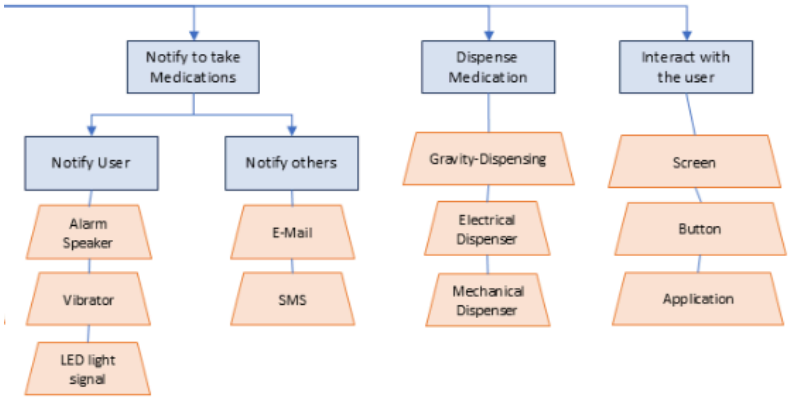
Function Means tree

The function-means tree takes in the previously identified functions and organizes potential means to complete each function or task.



(ZOOM IN)





Morphological Chart & Conceptual Designs

Means feature/function					
Control Humidi	rubber seal	desiccant materials	Humidity regulation system		
Preserving con	food-grad e plastic	stainless steel	glass		
prevent mix-up	Modular compart ments	Revolving compart ment	seperate d compart ments		
restock lid	large grip hinge lid	button triggerd	slide lid		
lid visibility	contrast color	contrast texture	LED		
safety access	Passcode	facial recognitio n	Fingerpri nt Scanner		
schedule meds	remote control	Control applicatio n	onboard control	AI	internal clock
properly dose	dosage calculator	AI	premeas ured compart ments		
Notify user	Alarm Speaker	Vibrator	LED light		
Notify others	E-mail	SMS	software		
Dispense medi	gravity dispensin g	electrical dispenser	mechanic al dispenser		
User interface	screen	buttons	app		

Design #1

- **Control Humidity : Rubber Seal** - Cheap, can get the job done.
- **Preserving Container : Food-Grade Plastic**
  Safe for medications, simple and lightweight.
- **Prevent Mix- up : Fixed Separated Compartment**
  Cheap and Simple way to prevent mix-up.
- **Restock Lid : Electronic Button Trigger**
  Control by electronics, easy to use with barely any effort.
- **Lid Visibility : LED** - Locate the location of the lid with light.
- **Safety Access : Passcode** - Access using PIN.
- **Schedule Medication : On-Board Control**
  controlled by On-board control.
- **Properly Dose Medication : Pre-Measured Compartment**
  Simple way to dose the right amount of medication.
- **Notify User : Alarm Speaker**
  Use noise to notify the user.
- **Notify Other : EMAIL**
  Use Email to notify the caregivers or relatives in case of emergency or error.
- **Dispense Medication : Mechanical Dispenser**
  Use a mechanical system to dispense the medications.
  User Interface : Buttons - Easy to use interface, basic and simple.

Design #2

- **Control Humidity : Humidity Regulation System** - Use an electronic system to regulate humidity.
- **Preserving Container : Stainless Steel** - Durable and safe for medications.
- **Prevent Mix- up : Modular Compartments** - Dynamic in the amount of medications can be stored. Easy to manage.
- **Restock Lid : Slide Lid** - Simple mechanic of a lid. Easy to comprehend.
- **Lid Visibility : Contrast Texture** - Using surface texture indicates the location of the lid.
- **Safety Access : Fingerprint Scanner** - Grant access to the patients only, using fingerprint recognition.
- **Schedule Medication : Control Application** - Control the pill box on a mobile phone application.
- **Properly Dose Medication : AI** - Using AI to learn the prescription and calculate the correct amount of medication.
- **Notify User : Alarm Speaker & LED** - Notify the user to take medication using light and sound.
- **Notify Other : SMS & Software** - Use some applications and SMS messages to notify the caregivers any error or emergency.
- **Dispense Medication : Electrical Dispenser** - Electrical system to dispense medication.
- **User Interface : Screen** - Use the screen user interface to interact with the user.

Design #3

- **Control Humidity : Desiccant Material -** Control Humidity using desiccant material like silica gel, etc.
- **Preserving Container : Glass**- Make the medication visible to the patient.
- **Prevent Mix- up : Revolving Compartment** - Revolving compartment for ease of use and selection.
- **Restock Lid : Large Grip Hinge** - Hinge lid with large grip/handle for ease of use in seniors.
- **Lid Visibility : LED** - Indicate the lid's location using light.
- **Safety Access : Passcode** - Access using PIN.
- **Schedule Medication : Internal mechanical Clock** - Use a built -in clock as a timer.
- **Properly Dose Medication : Pre-Measured Compartment** -- Simple way to dose the right amount of medication.
- **Notify User : LED** - Use lights to notify the user to take medication.
- **Notify Other : EMAIL -** Use Email to notify the caregivers or relatives in case of emergency or error
- **Dispense Medication : Gravity Dispensing** - Use Gravity to assist in medication dispensing.
- **User Interface : Screen** - Use the screen user interface to interact with the user.

Design #4

- **Control Humidity : Rubber Seal -** Affordable, can get the job done.

- **Preserving Container : Food-Grade Plastic** - Safe for medications, simple and lightweight.
- **Prevent Mix- up : Revolving Compartment** - Revolving compartment for ease of use and selection.
- **Restock Lid : Slide Lid** - Simple mechanic of a lid. Easy to comprehend.
- **Lid Visibility : Contrast Color & Texture** - Use color and surface texture to indicate the location of the lid.
- **Safety Access : Facial Recognition** - Use facial identity to grant access to the storage.
- **Schedule Medication : AI** - Using a well-trained AI to learn the prescription and schedule the time for medication.
- **Properly Dose Medication : AI** - Using AI to learn the prescription and calculate the correct amount of medication.
- **Notify User : Alarm Speaker -** Use voice and sound to notify the user.
- **Notify Other : SMS** - Use SMS to notify the caregiver in case of emergency or error.
- **Dispense Medication: Electrical dispenser** - Use an electric system to dispense medication.
- **User Interface: Screen** - Use screen for interaction.

Design #5

- **Control Humidity : Humidity regulation system** - Use an electric system to regulate humidity.
- **Preserving Container : Food-Grade Plastic** - Safe for medications, simple and lightweight.
- **Prevent Mix- up : Revolving Compartment** - Revolving compartment for ease of use and selection.
- **Restock Lid : Electronic Button Trigger** - Use an electric system to operate the lid.
- **Lid Visibility : Contrast Color & Texture** - Use color and surface texture to indicate the location of the lid.
- **Safety Access : Fingerprint Scanner**- Use fingerprint identity to grant access to the storage.
- **Schedule Medication : Control Application** - Use a mobile application to control and schedule medication according to prescription.
- **Properly Dose Medication : Dosage Calculator** - Use an electronic dosage calculator to calculate the dosage of the medication.
- **Notify User : Alarm Speaker -** Use voice and sound to notify the user.
- **Notify Other : SMS & Email** - Use SMS and Email to notify the caregiver in case of emergency or error.
- **Dispense Medication: Electrical dispenser** - Use an electric system to dispense medication.
- **User Interface: Screen** - Use screen for interaction.

Design Specification

1. The device must be able to store at least 3 different types of medications for varying use.
2. The user must be able to use the device within 3 - 4 minutes to be considered as “Easy to Use”
3. The device must not be longer and wider than 4 Centimeters and not thicker than 2.5 centimeters for it to be small and compact.
4. The device must not be weight must not be over 1.5 kilogram for the senior to be able to carry it.
5. The device must be able to notify the caregiver for any case of emergency or error as soon as possible.
6. The device must be able to preserve medication for at least 3 - 4 weeks.
7. The user interface of the device must be easy to use by the seniors. It should not take them more than 5 minutes to use the device.
8. The notifying system must be able to notify the senior patient to take their medication effectively. The user must be able to notice the notification within 1 - 5 minutes.
9. The time for restock should not take more than 3 minutes for the device to be easy to restock.
10. It should not take more than 5 minutes for the user to clean the device for the device to be considered as easy to clean.

Design Selection

Metrics

The metrics used are pulled from our design objectives list, each design is given a score ranging from 0 to 1 for each metric. The ratings are then used in the numerical decision matrix and design selection.

Easy to use

Scale:	Qualitative description	Range (Time)
1	Impossible	> 4 minutes
2	Nearly Impossible	>=3 minutes < 4 minutes
3	Possible	>= 2 minutes, < 3 minutes
4	Comfortable	>= 1 minute, < 2 minutes
5	incredibly easy	< 1 minute

ease of restocking

Scale:	Qualitative description	Range (Time)
1	impossible	> 4 minutes
2	Nearly Impossible	>=3 minutes < 4 minutes
3	Possible	>= 2 minutes, < 3 minutes
4	Comfortable	>= 1 minute, < 2 minutes
5	incredibly easy	< 1 minute

Size

Scale:	Qualitative description	Range (Base Area)
1	Very Big	> 6 cm x > 6 cm
2	Big	5 - 6 cm x 5 - 6 xm
3	Medium Size	4 - 5 cm x 4 - 5 cm
4	Small	3 - 4 cm x 3 - 4 cm
5	Compact	< 3 cm x < 3 cm

Affordability:

Scale:	Qualitative description	Range (CAD)
1	Costly	>= 40 CAD
2	Expensive	>= 30 CAD, < 40 CAD
3	Affordable	>= 20CAD, < 30 CAD

4	In-expensive	>= 10 CAD , < 20 CAD
5	Cheap	< 10 CAD

Communication:

Scale:	Qualitative description	Range (Time)
1	basically silent mode	>30minutes or none
2	forgettable	10-20 minutes
3	noticable	5-10 minutes
4	unable to ignore	1-5 minutes
5	very prominent	Within 1 minute

Medication volume and storage

Scale:	Qualitative description	Range (Time)
1	short restock period	>1 week
2	small restock period	2 week>.>1 week
3	ok period	3 weeks - 4 weeks
4	long time	4 weeks-5 weeks
5	very large restock period	>5 weeks

Number of possible medications

Scale:	Qualitative description	Range (# of different meds)
1	very restricitve	<=2
2	restricitve	<3
3	passable	3
4	versatile	>4
5	more than enough	>7

Easy to clean

Scale:	Qualitative description	Range (Time)
1	Sisyphus	< 3 minutes
2	difficult	>= 3 minutes, < 5 minutes
3	cleanable	>= 5 minutes, < 7 minutes
4	little effort	>= 7 minutes, < 9 minutes
5	no effort	>= 9 minutes

Numerical Evaluation Matrix

Design Objective/Contraints	Weight (%)	Design 1	Design 1 Score	Design 2	Design 2 Score	Design 3	Design 3 score	Design 4	Design 4 score	Design 5	Design 5 Score
Easy to Restock	7	0.9	6.3	0.5	3.5	0.8	5.6	0.5	3.5	0.8	5.6
Ease of Use	18	0.7	12.6	0.8	14.4	0.2	3.6	0.8	14.4	0.5	9
Size	4	0.5	2	1	4	0.5	2	0.5	2	0.5	2
Affordability	11	0.5	5.5	0.2	2.2	0.5	5.5	0.5	5.5	0.3	3.3
Communication	25	0.85	21.25	0.8	20	0.5	12.5	0.8	20	0.8	20
Volume of Medications	11	0.3	3.3	0.8	8.8	0.8	8.8	0.5	5.5	0.5	5.5
Variety of Medications	20	0.5	10	0.8	16	0.5	10	0.5	10	0.4	8
Easy to Clean	4	0.2	0.8	0.2	0.8	0.8	3.2	0.9	3.6	0.5	2
Totals	100		61.75		69.7		51.2		64.5		55.4

The following matrix shows that design 2 scored highest out of the five designs.

Design description

Design 2 was selected as our design as it is scored the highest in the numerical evaluation Matrix. Design 2 has its own unique characteristics and features which are:

Control humidity: A built-in humidity regulation system ensures medications remain in optimal conditions.

Preserving container: Made with stainless steel to protect medications and improve durability.

Modular compartments: These help prevent mix-ups and allow organized storage.

Restock lid: A sliding lid makes restocking simple and hassle-free.

Lid visibility: Designed with a contrasting texture for easy identification.

Safety access: A fingerprint scanner ensures only authorized users can access the medications.

Medication scheduling: Controlled via a dedicated app that simplifies reminders and dosage tracking.

Notification systems: Alerts include an alarm speaker, LED lights, and even SMS notifications to caregivers.

Electrical dispenser: Automated medication dispensing for accuracy and convenience.

User interface: A screen and app-based interface provide an intuitive user experience.

Project Planning:

Linear responsibility chart:

	Podcharakorn Sreeta	Ashton Neupane	Min Wai	Jeyoon Lee
Problem definition:(1.0)	2	1	3	3
(1.1) Clarify project	5	1	3	3
(1.1.1)Communication with client	4	2		
(1.1.2)assessing client statement	4	1	3	3
(1.1.3)Defining need	4	1	2	3
(1.1.4)Further client questioning	4	1		
(1.1.5)revised need	2	1	4	4
(1.1.6)defining goal	6	1	2	2
(1.2) Research	5	3	2	1
(1.2.1)Problem and causes research				1
(1.2.2)Existing designs research			1	
(1.2.3) Consumer requirements			1	
(1.3)Objectives and constraints	5	1	3	2
(1.3.1)Identify objectives and constraints			3	2
(1.3.2)Pairwise comparison		3	1	2
(1.3.3)Weighted Criteria tree			1	2
(1.3.4)Review with client	2	1		
Project Planning:(2.0)	5	1		
(2.1) Organizing	5	1	4	4
(2.1.1) Break down work structure.	4	1	3	3
(2.1.2) linear responsibility chart.	4	1		
(2.2) Scheduling	5	1		
(2.2.1) Form a work schedule using Gantt Chart	5	1		
Concept Generation:(3.0)	5	3	1	2
(3.1) Brainstorming Solutions	5	4	1	2
(3.1.1) Generate initial ideas for potential solutions			1	2
(3.1.2) Research similar products and gather insights for inspiration			2	1
(3.2) Concept Analysis	5	3	1	2
(3.2.1) Identify the pros and cons of each proposed solution		3	2	1

(3.2.2) Evaluate designs for feasibility, usability and potential impact	1	6		
(3.3) Concept Refinement	5	6	1	2
(3.3.1) Narrow down to the three concepts based on analysis			1	2
(3.3.2) collect census from potential consumers		1	3	
Design Selection:(4)	5	3	3	2
(4.1) Select a design concept	5	1	3	2
(4.1.1) Identify the principal attributes		3	1	2
(4.2) Numerical decision matrix	5		1	2
(4.2.1) Method for selecting particular designs			2	1
(4.2.2) List of design criteria	5	3	2	1
(4.2.3) Rank importance of criteria	5	6	2	1
(4.2.4) Scales of the degrees in criteria			2	1
(4.2.5) Design concepts	5		2	1
(4.3) Matrices	5	3	1	2
(4.3.1) Design objectives		4	2	1
(4.3.2) Scale to measure design performance			1	2
(4.3.3) Identify units and scale for measurement		4	2	1
(4.3.4) Identify a method of assessment	3	2	1	2
(4.4) Design selection	5	6	3	2
(4.4.1) Material Selection.			1	
(4.4.2) Create multiple designs to be considered.		4	2	1
(4.4.2) consider the best design.	2	1		
Modeling and Analysis:(5.0)	5	4	2	1
(5.1) Model Chosen Design	5		1	2
(5.1.1) Create a Blueprint of the device.			1	2
(5.1.2) Create software for the device.			1	2
(5.1.3) Create design parameters.	4		1	2
(5.1.4) 3D model to visualize design and simulate functionality.			1	2
(5.2) Analyze the chosen design	5	1	3	4
(5.2.1) Evaluate and verify that the prototype meets its specifications.	1	2	3	4
(5.2.2) Review the design to make it meets its design objectives.	1	3	2	4
Implementation:(6.0)	5	3	1	2
(6.1) Testing the design	5		2	1
(6.1.1) Create an evaluated prototype.			1	2

(6.1.2) Verify that the prototype meets its design objectives.		1	1	2
(6.1.3) Iterations and rework on the prototype.	1	4	2	3
(6.1.3.1) Review the design of the hardware to meet its objectives.	1	6	3	2
(6.1.3.2) Review the design of the software to meet its objectives.	1	6	3	2
(6.2) Detailed Design	5	2	1	2
(6.2.1) Rework on the prototype to make it follow the laws and regulations.	4	4	1	3
(6.2.2) Optimize and refine both software and the hardware.			1	3
(6.2.3) Public hearing/ get feedback from the public and clients.	2	1	5	
(6.2.4) Implement the feedback and rework on the device and software.	4	6	1	3
(6.3) Submit the final design for the client to review.	2	1		
Final Documentation:(7.0)	5	1	3	4
(7.1)Design papers	5	1	3	
(7.1.1)review design blueprints	1	2	2	2
(7.1.2)finalize blueprints	4	6	1	2
(7.1.3)Finalize manufacturing costs	4	1		
(7.1.4)design justification	4	6	1	2
(7.2)Presentation	5	1		
(7.2.1)present design to client		1		
(7.3) Final Documentation	5	1		
(7.3.1)Review final documentation and finish project.	1	6		

Gantt Chart


Gantt Chart
Duration (Day(s))
September
October
November
December
Start Project 0
1.1 Clarify the project 7
1.2 Research 14
1.3 Objective and Constraints 7
2.1 Organizing 14
2.2 Scheduling 14
3.1 Brain Storming solutions 7
3.2 Concept Analysis 7
3.3 Concept Refinement 7
4.1 Select the design concept 7
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